

```
root@iitdu-forensics:~#
```

Digital Footprints: Linux Log Analysis for Digital Forensics

*A hands-on session on reading, interpreting, and investigating
Linux system & web-server logs after a security incident.*

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EWU Cybersecurity Club (EWUCSC)

Session at IITDU | Digital Forensics Track

WHO AM I??

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- Security Researcher | Blue Teamer | Penetration Tester
- Certified Network Security Practitioner (CNSP)
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Agenda

01

What Is Linux Log Analysis?

Foundations of forensic log reading

02

Web Server Logs & Cyberattacks

What they capture, and why they matter after a breach

03

Core Concepts of Logs

Sources, timestamps, levels, rotation & integrity

04

Essential Linux Commands

grep, awk, journalctl, and a live investigation walkthrough

05

Logs in Real Job Life

What SOC, DFIR & pentest careers actually look like

What Is Linux Log Analysis?

The systematic process of collecting, reading, and interpreting log files produced by the Linux kernel, system daemons, and applications to reconstruct events, detect anomalies, and answer: who did what, when, where, and how.

In forensics, logs are the closest thing to a **“black-box flight recorder”** for a server — every login, request, and privilege change leaves a trace.

```
analysis_goals.sh

$ detect_anomalies      # spot unusual behavior

$ reconstruct_timeline  # order events by time

$ identify_ioc          # IPs, hashes, payloads

$ trace_intrusion       # entry -> lateral ->
exit

$ preserve_evidence     # for reporting / court
```

What Are Web Server Logs?

Every request to a web server (Apache, Nginx) is automatically recorded — forming a complete, timestamped record of who accessed what.

Access Log

Records every HTTP request: client IP, timestamp, requested URL, HTTP method, status code, user-agent.

```
/var/log/apache2/access.log  
/var/log/nginx/access.log
```

Error Log

Captures server-side failures: crashed scripts, permission denials, misconfigurations, module errors.

```
/var/log/apache2/error.log  
/var/log/nginx/error.log
```

Sample access-log line: `203.0.113.7 - - [19/Aug/2026:10:22:41] "GET /admin.php HTTP/1.1" 404 512`

Why Logs Matter After a Cyberattack

1

Reconstruct the Attack Timeline

Trace the intruder's path: reconnaissance -> exploitation -> persistence -> exfiltration.

2

Identify Indicators of Compromise

Extract attacker IPs, malicious payloads, suspicious user-agents, and injected commands.

3

Determine Scope & Impact

Find exactly which files, accounts, or endpoints were accessed or modified.

4

Support Legal & Compliance Evidence

Provide an auditable, time-stamped record for incident reports and chain of custody.

Core Concepts of Logs

Log Sources

Kernel, system daemons (systemd), applications, and network services each generate their own logs.

Timestamps

Every entry is time-stamped; accurate NTP sync is essential to build a reliable timeline.

Log Levels

DEBUG, INFO, WARNING, ERROR, CRITICAL — severity determines what needs urgent review.

Log Rotation

logrotate archives & compresses old logs so disks don't fill up, while preserving history.

Centralization

Forwarding logs to a SIEM (e.g. rsyslog, ELK, Splunk) for correlation across many hosts.

Integrity

Hashing & write-once storage protect logs from tampering — critical for evidence.

Where Linux Logs Live

ls -la /var/log/

/var/log/auth.log	Debian/Ubuntu	Authentication: logins, sudo use, SSH sessions
/var/log/secure	RHEL/CentOS	Authentication & authorization events
/var/log/syslog	Debian/Ubuntu	General system activity across services
/var/log/messages	RHEL/CentOS	General system & kernel messages
/var/log/kern.log	Most distros	Kernel-level events & hardware errors
/var/log/apache2 nginx/	Web servers	access.log & error.log for HTTP traffic
journalctl	systemd (all)	Unified binary journal for every service

Essential Linux Commands for Log Analysis

```
$ tail -f /var/log/auth.log
```

Live-follow a log as new events arrive

```
$ grep "Failed password" auth.log
```

Filter lines matching a pattern

```
$ awk '{print $1, $9}' access.log
```

Extract & format specific fields

```
$ sed -n '100,200p' syslog
```

Print a specific range of lines

```
$ journalctl -u sshd --since "1 hour ago"
```

Query the systemd journal by service & time

```
$ last -a
```

Show recent login history with source host

```
$ sort | uniq -c | sort -nr
```

Count & rank repeated values (e.g. top IPs)

```
$ zgrep "error" access.log*.gz
```

Search inside rotated/compressed logs

Live Example: Spotting a Brute-Force Attack

Goal: find which IP address made the most failed SSH login attempts.

```
investigation.sh

$ grep "Failed password" /var/log/auth.log \
  | awk '{print $(NF-3)}' \
  | sort | uniq -c | sort -nr | head -5
```

```
142 198.51.100.23
37 203.0.113.7
9 192.0.2.44
```

Reading the result: 198.51.100.23 attempted 142 failed logins — a strong brute-force indicator. Cross-check this IP against threat-intel feeds and firewall logs next.

Output of Log Analysis in Job Life



SOC Analyst

Monitors logs in real time to detect and triage incidents before they escalate.



DFIR Investigator

Reconstructs attacker timelines from logs to produce court-ready forensic reports.



Penetration Tester

Reviews logs to understand detection coverage and improve red-team tradecraft.



DevSecOps Engineer

Builds logging & alerting pipelines that keep infrastructure auditable and compliant.

Bottom line: Log analysis is one of the most consistently in-demand, hands-on skills across every cybersecurity career path — and it's tested directly in interviews, CTFs, and certifications.

Key Takeaways



Logs are the DNA of digital forensics — every action leaves a trace.



Know your core log locations: auth.log, syslog, web server logs, journalctl.



Master grep, awk, sed, and journalctl — they solve 90% of real investigations.



Always correlate timestamps across sources to build one reliable timeline.



Practice on real logs: CTFs, home labs, and open datasets build true skill.

```
root@iitdu-forensics:~# whoami
```

Thank You

Questions & Discussion

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